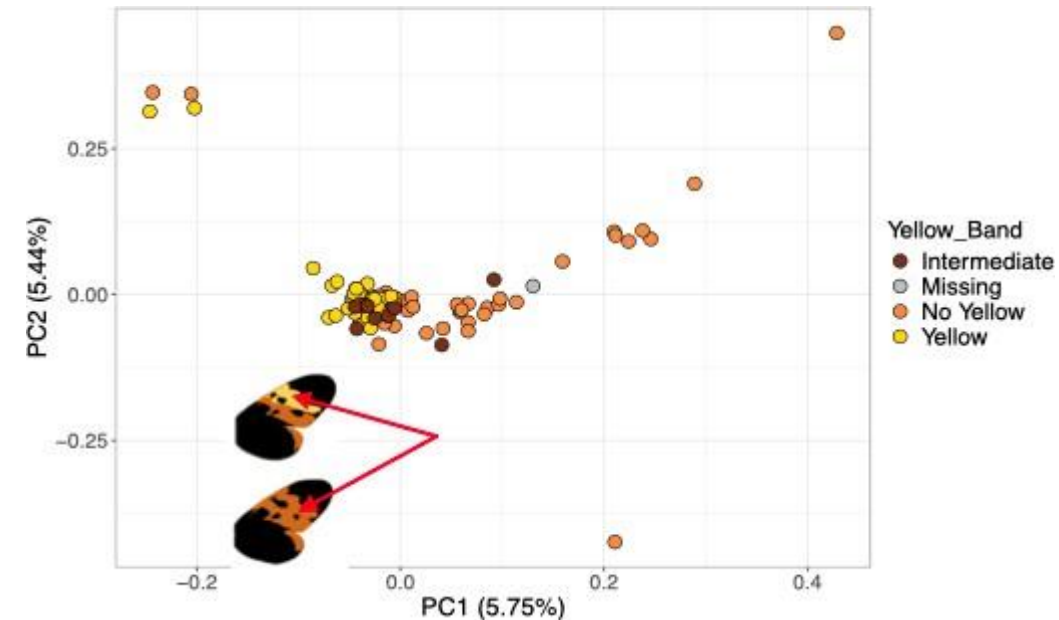


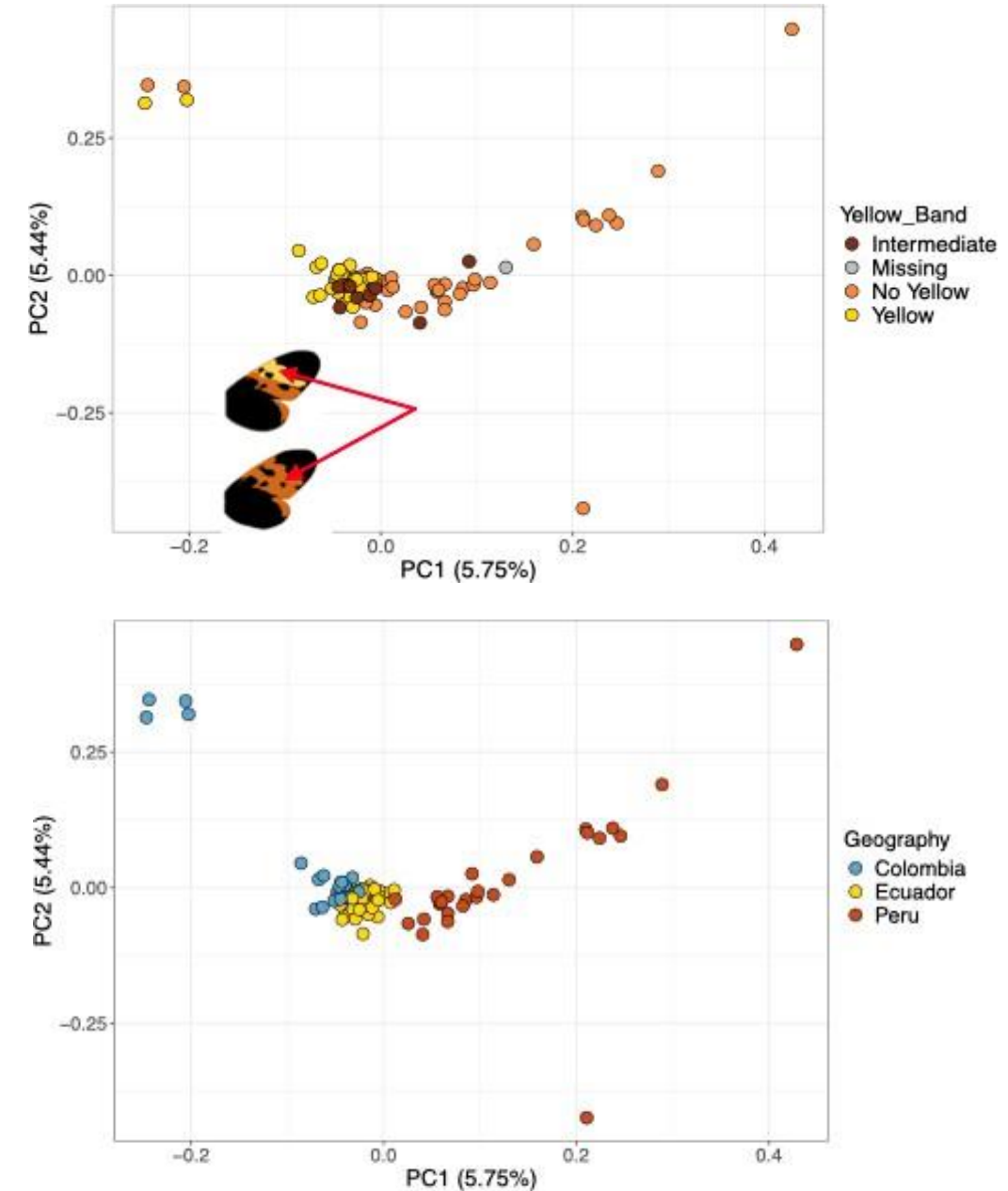
PCA – Principal Components Analysis

- **Complexity reduction:** PCA reduces the dimensionality of the data
- PCA identifies the **main axes of variation** in a dataset with each axis being independent of the next
- **Eigenvalues:** each axis has an eigenvalue, that can be transformed into the percentage of variance explained
- **Eigenvectors:** each individual has a position on each axis



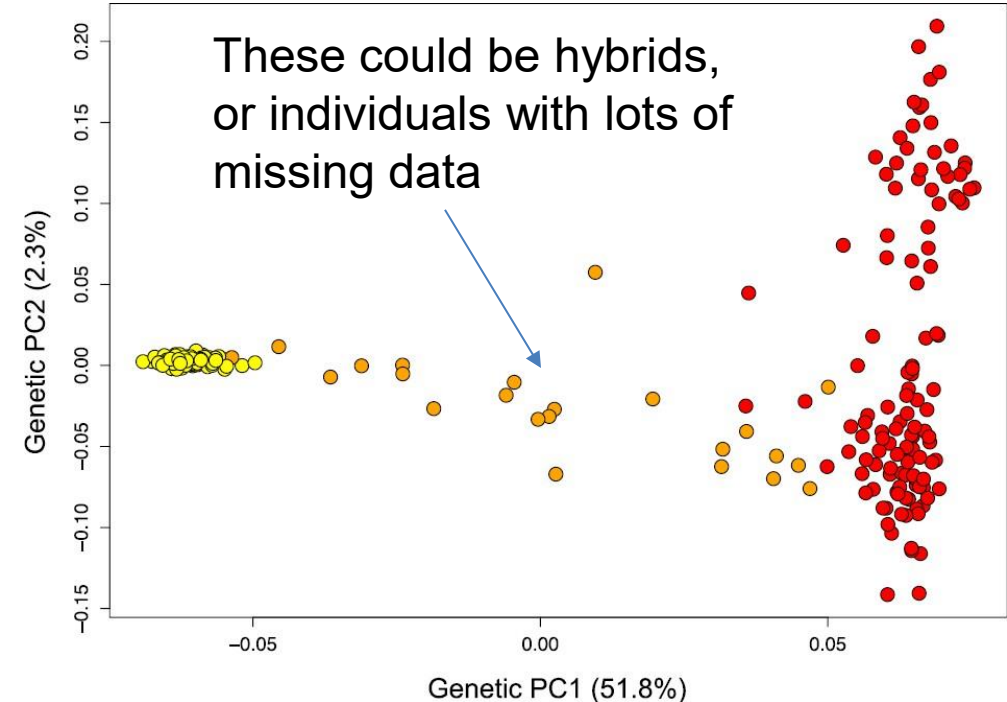
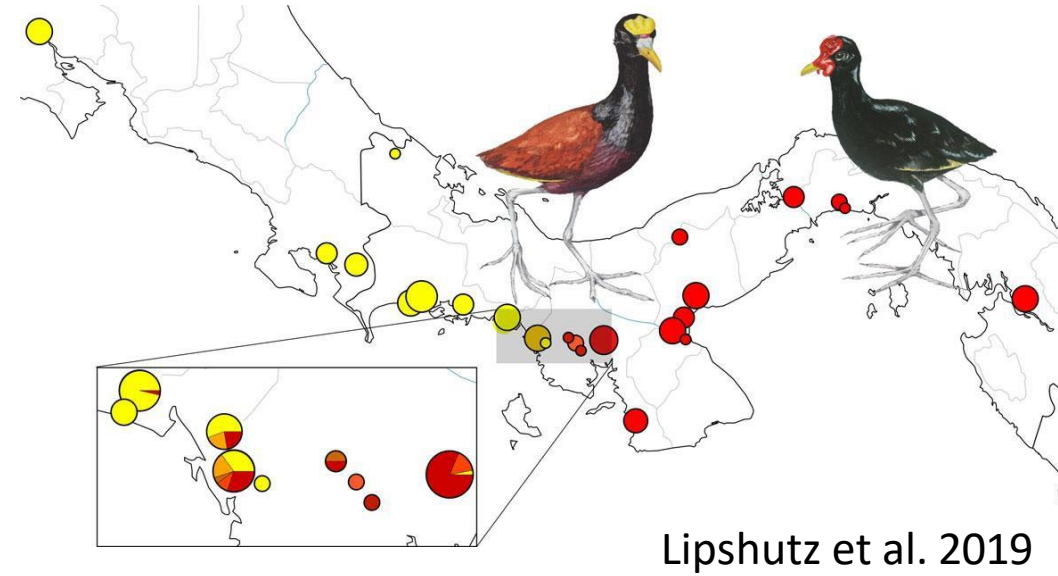
Why run a PCA?

- Are all individuals clustering together as expected? (e.g. no outlier or misidentified individuals)
- Testing if individuals with different colour, morphology or geographic origin cluster separately.
- Are any individuals genetically intermediate?



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Assumptions of PCA

- All sites are independent
- For genomic data:
only SNPs (no insertions or deletions)
- Individuals are closely related and populations are more or less evenly represented.

If you have only few individuals each of divergent species
-> better use phylogenetic approaches